

Mental Accounting Through Two-stage Budgeting Under Bounded Rationality

Nick Pretnar

npretnar@ucsb.edu

University of California, Santa Barbara

Christopher Y. Olivola

Indiana University

Alan Montgomery

Carnegie Mellon University

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Abstract

We propose a model that generalizes two-stage budgeting to incorporate frictions which induce budgetary stickiness and mental accounting behavior. Inattention limits the fungibility of money across mental accounts, leading to boundedly rational behavior. Financial attentiveness is cognitively costly, so consumers might re-assess only a subset of spending budgets every period. Over- or under-spending affects future budgeting and expenditure decisions. Using agent-level weekly expenditure data, we structurally estimate how much low-income consumers appear rationally constrained in budgeting, providing insights into how consumers would respond to interventions that encourage budgeting discipline. Attentiveness constraints can have a disciplinary effect on finances: relaxing constraints leads some consumers to overspend and become financially worse off.

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1 Introduction

Budgeting is often touted as a way for consumers to discipline their expenditure decisions. However, there is still much we do not know about how consumers use or adhere to budgets in their everyday lives. Mental accounting theory, in part, posits that consumers use budgets to discipline their spending by tagging money for specific purposes (Thaler, 1999). Meanwhile, in classical expenditure models money is assumed perfectly fungible and freely transferable (Deaton and Muellbauer, 1980a). Our goal is to build a general model of spending and budgeting behavior that allows for individual agents to exhibit varying degrees of bounded rationality. We then estimate this micro-founded model of consumer behavior on agent-level expenditure data to quantify the degree to which agents' expenditure patterns reflect mental accounting behavior.

We establish that consumers exhibit rich heterogeneity with respect to the degree to which their budgeting appears boundedly rational. We define a pure mental accounter as one who does not rebalance their budgets but abides by the dictates of their initial allocation. Our quantitative results indicate that consumer decision-making is neither fully rational nor fully irrational, but rather budgeting behavior falls along a rationality continuum. In our set-up pure mental accounting and perfect fungibility are opposing, extreme cases of a fungibility continuum. Our approach nests multiple features from the behavioral economics literature such as: 1) fungibility frictions that result from timing constraints (i.e., ex-ante budgets are only equal to ex-post expenditure on a measure-zero set of expenditure/preference shock realizations); 2) narrow choice bracketing as in Gabaix (2014) and Lian (2021) (i.e., consumers may update some budgets for certain expenditure categories in one period while leaving other budgets unchanged); 3) numeracy constraints with respect to setting optimal budgets (i.e., consumers may only update a budget if the optimal candidate budget is significantly different from the underlying budget with which they enter the period); 4) mental accounting (i.e., future budgets may be informed by past behavior with respect to over- and under-expenditure relative to past budgets).

To test the degree to which each of the different behavioral features are important for predicting high-frequency time series of expenditure, we estimate the model using weekly, agent-level expenditure data for a set of low-income consumers. Our quantitative strategy is designed to answer several questions through the model's lens. First, does empirical evidence suggest that

consumers use budgets to discipline and constrain their expenditure decisions? If so, how do the ways in which people use budgets vary? Do consumers regularly revise all of their budgets or only a subset? Finally, what are the implications for both welfare and savings rates when such rigidities are either enhanced or relaxed?

The model, while built with theory and consumer behavior in mind, is primarily structured so as to be able to quantitatively infer agent-level budgeting and mental accounting behavior while only observing expenditure. We show that we can invert the model’s first-order conditions in order to tractably recover latent budgeting decision variables conditional upon observed expenditure data (and, of course, the theoretical structure of the model). Though, due to space constraints, we do not discuss the details of our estimation routine in the main text, one major technical contribution we provide is the tractable (but probabilistic) inference of latent behavioral objects underlying observed expenditure. Our quantitative results are thus “model dependent,” but this is necessary if we are to actually push forward the frontier of knowledge with regards to consumer behavior, which heretofore has primarily been investigated via a combination of theory and lab experiments, with only narrow applications to actual field data.¹

Our best-estimated models predict that consumers optimally re-evaluate budgets for approximately half of their consumption categories in any given period. However, there is substantial heterogeneity across consumers with respect to how they use budgets and how they respond to expenditure/preference shocks. A plurality of consumers use mental accounting and sticky budgets to discipline their total spending and thus savings behavior. A substantial subset of these consumers exhibit a fascinating, albeit financially perplexing, pattern of behavior: they maintain spending discipline by updating their budgets *infrequently* due to the cognitive costs of doing so. We predict that for these consumers, lowering the cognitive costs of updating their budgets may push them to raise their budgets to avoid the dis-utility of over-spending, yielding a repetitive budgeting/spending cycle that ultimately leads to debt accumulation and, in extreme cases, bankruptcy.

The paper proceeds as follows. In §2 we position our work in the broader literature of two-stage budgeting, bounded rationality, and mental accounting. §3 builds a model of two-stage

¹See, for example, [Hastings and Shapiro \(2018\)](#), who quantitatively analyze SNAP (Supplemental Nutrition Assistance Program) benefit utilization for evidence of mental accounting behavior.

budgeting that incorporates features from behavioral economics. §4 describes our unique dataset of consumer-level weekly spending and income. This dataset motivates our underlying model construction from §3 since we want to allow for endogenous variation and heterogeneity in observed weekly spending patterns at the consumer level. §5 estimates the model and discusses the results of our estimation and their implications for inference with regards to mental accounting theory, budgeting, and cognitive/financial attentiveness, including as pertaining to numeracy constraints. We conclude with a discussion in §6 of our findings and limitations of this research.

2 Motivating Literature

2.1 Two-stage Budgeting and Classical Demand

The theory of two-stage budgeting, which posits that consumers first allocate expenditure shares to broad commodity categories prior to making individual spending decisions, provides micro-foundational justification for estimating demand systems derived from separable utility models. The idea behind two-stage budgeting can be traced back to theories of aggregation described independently in Hicks (1936) and Leontief (1936). Each author shows that if the marginal rate of substitution for two goods in commodity group j is independent of the marginal rate of substitution for two goods in group k , then preferences are at least weakly separable over those broad commodity groups.² Building on the two-stage budgeting framework, Deaton and Muellbauer (1980a) argue that if prices for commodities within a group move in parallel then weak separability can be assumed. The consumption decision process can then be framed around these broad aggregates, where consumers are assumed to allocate resources for consumption expenditure not by optimally responding to posted prices of individual commodities, but only by considering price levels for broad groups of them. Indeed, tractable demand systems, like the linear expenditure systems of Geary (1950), Stone (1954), and Houthakker (1960), the Rotterdam models of Barten (1964), Theil (1965), Barten (1967), Theil (1976), and Barten (1977), or the almost ideal demand system of Deaton and Muellbauer (1980b), are generated from a utility function with some degree of separability, exploiting these aggregation results.

²Gorman (1959) (later extended by Blackorby and Russell (1997)) recognized the importance of this result in order to generate preference structures that aggregate.

Such models implicitly rely on two strong assumptions about consumer behavior. First, consumers are assumed to exhibit perfect fungibility; that is, they can freely transfer money that has been allocated to different commodity-group budgets. Second, first-stage budgets and second-stage spending are assumed to always equate, so that there are no timing frictions whereby consumers engage in a level of spending that violates their ex-ante plans. Our contribution is to consider both theoretical and empirical results from a model environment in which these assumptions do not necessarily hold. Hence, our framework relaxes the traditional two-stage budgeting assumptions, where these relaxations can be interpreted as allowing for mental accounting behavior.

2.2 Bounded Rationality and Rational Inattention

Among our additions to the two-stage budgeting framework, we allow for heterogeneous sparse maximization, similar to the setup in [Gabaix \(2014\)](#). This places our research into conversation with contemporary economics research, popularized by [Sims \(2003\)](#) and thoroughly reviewed in [Maćkowiak et al. \(2022\)](#), where consumers make decisions under limited information or with limited cognitive resources to optimize. This feature of our framework follows from the well-established finding in cognitive psychology that people can only attend to, and process, a limited amount of information at any given time ([Miller, 1956](#); [Simon, 1957](#); [Cowan, 2000](#)). In our sparse-max framework, consumers might only re-evaluate a subset of their first-stage budgets in any given period, thereby engaging in narrow choice bracketing ([Kahneman and Lovallo, 1993](#); [Read et al., 1999](#); [Rabin and Weizsäcker, 2009](#); [Felső and Soetevent, 2014](#); [Koch and Nafziger, 2016, 2019](#)). By allowing for heterogeneous rational inattention and heterogeneous non-fungibility in this way, our approach reflects a broad literature which suggests there is no reason to assume that all consumers will regard all attributes of a decision problem as uniformly salient to the same degree ([Chetty et al., 2009](#); [Sussman and Olivola, 2011](#); [Kőszegi and Szeidl, 2013](#); [Bordalo et al., 2014](#); [Schwartzstein, 2014](#); [Caplin and Dean, 2015](#); [Abeler and Marklein, 2017](#); [Read et al., 2017](#); [Bordalo et al., 2020](#); [Kőszegi and Matějka, 2020](#); [Bordalo et al., 2022](#)). Under such a structure, we will use our model to show empirically that different consumers appear rationally constrained to varying

degrees over different time intervals.³

It is important to note that we are not the first to consider how choice bracketing, budgeting, and substitutability are linked. [Kőszegi and Matějka \(2020\)](#) consider a similar, theoretical environment to ours, where high attention costs lead to essentially fixed budgets. A key result in their model predicts that broad budgeting behavior depends on the underlying substitutability of the commodities, so that highly-substitutable commodities are more likely to be budgeted together. This is an inherent result of a consumer’s inattentiveness: they lack the intrinsic ability to think about how to optimally distribute resources amongst commodities each of which is substitutable with another to varying degrees. Thus, the model in [Kőszegi and Matějka \(2020\)](#), using behavioral insights, yields the classical aggregation result in [Hicks \(1936\)](#), [Leontief \(1936\)](#), and [Deaton and Muellbauer \(1980a\)](#) — that separability is supported if the elemental commodities that comprise broad commodity-groupings are highly substitutable.

2.3 Mental Accounting and Budgeting Commitment

Our approach to modeling mental accounting builds on, but also departs from, prior work. On one hand, we motivate our model by including budgeting behavior consistent with some aspects of the planner/doer models of [Shefrin and Thaler \(1981\)](#) and [Shefrin \(2020\)](#), as well as the dual-self, budget commitment model of [Galperti \(2019\)](#) and the multi-self, mental accounting model of [Lian \(2021\)](#). We depart from much of the existing mental accounting literature in that we seek to explain behavioral tendencies not as resulting from reference-dependent utility or price adaptation, such as the model in [Baucells and Hwang \(2017\)](#), but rather generated by bounds on cognitive attention and numeracy. Thus, our approach is most similar to [Lian \(2021\)](#) who considers how narrow thinking can lead to outcomes that resemble narrow choice bracketing, though we build our model with an explicit aim toward estimation on high-frequency expenditure data as opposed to theoretical exposition. In our framework a boundedly rational consumer may only have the cognitive attention to re-evaluate and, if needed, update a subset of their first-stage expenditure budgets in a given period. In our model, mental accounting is just a book-keeping mechanism

³As an aside, we should note that we are not the only recent working paper to find heterogeneity in the degree to which consumers exhibit rationality in their consumption decisions. We will later show that the rationality heterogeneity implied by our model is broadly consistent with recent findings on consumption responsiveness to small and large income shocks discussed in [Andre et al. \(2025\)](#).

which may inform budget responsiveness: over-spending in period t could lead to reduced or increased budgets in period $t + 1$ depending both on the consumer’s disutility of over-spending and their marginal propensity to save. The so-called ex-ante planner may take into consideration relative over- or under-expenditure in multiple different commodity groups when making a narrow decision regarding their budget for one particular category, only. In this way, we introduce the accounting mechanism from [Thaler \(1985\)](#) that dynamically keeps track of over- or under-spending, into a model environment where the breadth of choice bracketing determines the degree to which over- or under-spending actually affects future budgets.

This interaction governing the degree to which rationality (i.e., budget attentiveness) is bounded and over- or under-spending impact future budgets is what we interpret as mental accounting. As [Farhi and Gabaix \(2020\)](#) point out, however, the literature contains no widely agreed-upon definition as to what exactly constitutes mental accounting. For example, Thaler uses the term to describe different behaviors, such as keeping track both of spending on certain consumption items and, separately, spending using specific (different) liquidity sources ([Thaler, 1985](#); [Thaler et al., 1997](#)). In this paper, we abstract from method of payment mental accounting dependencies as in [Feinberg \(1986\)](#), [Prelec and Loewenstein \(1998\)](#), and [Mullainathan \(2002\)](#). This is not a problematic abstraction for us because the data we use to validate the model are from pre-paid debit card users who are underbanked and most likely to use their card as their dominant liquidity source. Since low-income consumers are more likely to be liquidity-constrained and liquidity-constrained households are budget-sensitive and often engage in non-fungible spending behavior for basic necessities like food ([Gelman et al., 2014](#); [Hastings and Shapiro, 2018](#)), studying the spending patterns of low-income households is a natural application for a model with endogenous mental accounting and budgeting features. Our empirical approach follows after recent theoretical work in [Kőszegi and Matějka \(2020\)](#), who show that category-specific budgets are useful for consumers with high attention costs, many of whom are more likely to be low-income ([Schilbach et al., 2016](#)). Given our field data, we thus contribute to the mental accounting and budgeting literature by examining the degree to which low-income consumers, via their spending behavior, appear to exhibit traits consistent with contemporary behavioral theories at the frontier of present research.

3 Model

We construct the model in order that it may be tractably estimated with our dataset featuring only expenditure data,⁴ while still featuring as many behavioral features as we can possibly allow for. Specifically, in our data we observe individual, weekly-level expenditures from which we want to infer latent budgets. Since latent inference of underlying decisions is required, this necessitates a highly stylized model that can easily be mapped into a latent-inference algorithm to probabilistically uncover latent budgeting choices on which expenditure observations depend.

Time is discrete and indexed by t . Column vectors and matrices are denoted with a bold typeface. All variables and functions presented, except market prices $\mathbf{p}_t$ and gross interest rates r_t , are agent-specific, with consumer units indexed by i . Consumers make decisions in a sequentially dynamic environment, choosing savings so as to satisfy utility over liquidity holdings.⁵ A single period's decision process is divided into two sub-stages. At the beginning of each period consumers choose/plan their budgets subject to some restrictions to maximize their utility and end the period by realizing unanticipated random shocks to preferences and prices on which their actual expenditure depends.

In this sense our decision-theoretic setup has a planner/doer flavor a la [Shefrin and Thaler \(1981\)](#). At the beginning of each period in the first stage of the decision problem, the planner chooses optimal budgets subject to various cognitive constraints, which we will describe below; meanwhile, in the second stage, the doer reacts to some shock process that causes prices and preferences to deviate from their ex-ante expected values. The doer thus engages in expenditure which deviates from the budget/plan formed by the planner in the first stage: the doer may over- or under-spend, depending on the nature of the shock realization. Then, in the next period when the consumer is again making a planning decision, they do so internalizing the degree to which the doer over- or under-spent in the previous period. The amount of over- or under-spending by consumption category is thus the mental accounting state variable on which budgeting decisions

⁴We describe the dataset in more detail in [§4](#).

⁵Consumers have preferences for carrying forward liquidity (balances) and, via this mechanism, indirectly smooth consumption. In this way consumers do not solve for today's consumption as a function of a forecast of future consumption. This setup is strictly undertaken for tractability purposes: as will become apparent, the balance of liquidity carried forward acts as a sufficient statistic for summarizing the dynamics of latent decisions, which in our setup depend on this balance rather than on agents' rational forecasts of optimal (and to the econometrician, unknown) future consumption.

depend.

Behavioral frictions are incorporated into the model along several dimensions. First, consumers are assumed to face cognitive constraints regarding their budget evaluations: that is, in any given period, they may only have the cognitive capacity to re-evaluate some subset of their consumption budgets. Second, consumers imperfectly choose budgets so as to optimally set ex-ante expenditure expectations, though expenditure itself (either due to unanticipated price and/or utility shocks) will only ever exactly adhere to the ex-ante budget in measure-zero outcomes. Third, ex-ante budget re-evaluation is also assumed to be subject to a numeracy constraint: consumers may not necessarily make the effort to change a budget if the proposed, optimal budget is not significantly different from the budget already in place.

The equilibrium concept we deploy is that of the *personal equilibrium*, where decisions must be consistent with boundedly rational (i.e., constrained) optimization of some decision-theoretic, concave programming problem.⁶ In this sense we do not use the term “equilibrium” to describe a set of statistics that summarize market outcomes such as prices.⁷ Rather, we use the term to describe a set of decision-theoretic outcomes that summarize the deliberations undertaken in some kind of personal decision process where agents are subject to behavioral constraints and behave as if (at times) they are actually multiple, rational agents each engaging in different kinds of decisions. Specifically, our model formulation and the equilibrium concept associated with it will resemble the planner/doer formulation of [Shefrin and Thaler \(1981\)](#) in which a consumer can be thought to consist of two “selves” as in [Galperti \(2019\)](#) and [Lian \(2021\)](#) — 1) a planner who establishes ex-ante budgetary constraints; 2) a doer who engages in consumption expenditure subject to those constraints, which may or may not be associated with some kind of stochastic shock process. The planner can be thought of as forward-looking, forming budgets while thinking about the future, while the doer myopically reacts to various exogenous stimuli leading to expenditure deviations from the planner’s budgets.

We proceed in §3.1 by first describing the decision process of the myopic doer, who essentially just responds to various exogenous stimuli. In this section we also describe ex-post realized con-

⁶The term *personal equilibrium* is commonly deployed in behavioral economics to characterize decision-theoretic outcomes subject to various kinds of rationality constraints (see, for example, [Kőszegi \(2010\)](#) and [Freeman \(2017\)](#), along with discussions in [Thaler \(2016\)](#)).

⁷This is the common usage of the word in aggregate economics, for example.

sumption preferences and outline how actual liquidity balances evolve. In §3.2 we move on to describe the ingredients of the ex-ante problem faced by the planner. Note that each period the planner acts *before* the doer, anticipating the doer’s actions in that period subject to some set of behavioral constraints. The planner is thus the “first stage” actor in a classic two-stage budgeting model without behavioral frictions, while the doer is the “second stage” spender (Deaton and Muellbauer, 1980a). Unlike the doer, the planner is not myopic: they make plans in period t by acting on information with regards to the doer’s behavior in period $t - 1$, as summarized by a mental accounting state variable that keeps track of the doer’s over- or under-expenditure relative to last period’s budget. In §3.3 we characterize the decision problem faced by the first-stage planner when engaging in budget updates, including how numeracy constraints affect budgets. Finally, we conclude the model outline in §3.4 where we formally define the *personal mental accounting equilibrium*.

3.1 The Doer

In the spirit of Shefrin and Thaler (1981), the “doer” in our model is the aspect of the consumer that makes consumption expenditure decisions subject to exogenous and random deviations around a given budget. The doer realizes utility from consumption, while the planner forms budgets by trying to maximize expected utility subject to various behavioral constraints.

Utility. Consumers have direct utility defined over a J -dimensional vector of real quantities of consumption $\mathbf{q}_{it}$ and unallocated liquidity for period- t of $\bar{z}_{it}$. For tractability purposes we choose a strongly separable utility parameterization:⁸

$$u_i(\mathbf{q}_{it}, \bar{z}_{it}) = \sum_{j=1}^J \alpha_{ij} \ln(q_{ijt} + 1) + \alpha_{i,J+1} \ln(\bar{z}_{it}) \quad (1)$$

Flow utility from consumption has a Stone-Geary representation that also allows for zero consumption in a given period (Geary, 1950; Stone, 1954). Further, the consumer has preferences over

⁸Strong separability ensures we can analytically solve for the planner’s equilibrium budgeting decisions, which will become apparent in §3.3.

available liquidity for expenditure.⁹ We assume $\alpha_{ij} \in (0, 1)$ and $\sum_{j=1}^J \alpha_{ij} = 1$. After characterizing the income and savings processes, we will return to the utility function to discuss the role of $\alpha_{i,J+1}$.

Unallocated Liquidity from Borrowing, Savings, and Balances. The sum of marginal period- t savings z_{it} (flow of savings), the consumer's borrowing limit m_i , and available bank balances b_{it} is unallocated liquidity, $\bar{z}_{it} = z_{it} + m_i + r_t b_{it}$. Account balances evolve according to $b_{i,t+1} = r_t b_{it} + z_{it}$. The borrowing limit is such that $m_i > -b_{it}$ always, and returns are non-negative.¹⁰

Income and Savings. Let ℓ_{it} be period- t income and $x_{ijt} = p_{jt} q_{ijt}$ be expenditure on commodity group j . Note that marginal savings is $z_{it} = \ell_{it} - \sum_{j=1}^J x_{ijt}$. Savings can be negative ($z_{it} < 0$) if consumers spend more than they earn in income. After substituting the expression for z_{it} into $\bar{z}_{it}$ and then (1) it can be shown that the utility function is strictly increasing and strictly concave in q_{ijt} , thus leading to unique allocations, as long as m_i is sufficiently large, and $\alpha_{i,J+1}$ is strictly positive.¹¹

Expenditure Shocks. We model ex-post expenditure shocks that hit the consumer in the second-stage of the decision process as follows. Let ω_{ijt} be the budget inherited from the planner's first-stage constrained decision problem. We will discuss the structure of this object in more detail in §3.2 below. The doer takes this object as given, as if it is a fixed parameter. The doer realizes ζ_{ijt} , an *iid* idiosyncratic expenditure shock, which captures how much ex-post expenditure deviates from the ex-ante budget:

$$x_{ijt} = \omega_{ijt} + \zeta_{ijt}, \quad (2)$$

Given ω_{ijt} , (2) exactly characterizes expenditure, so that consumption is $q_{ijt} = (\omega_{ijt} + \zeta_{ijt})/p_{jt}$. Since the vector ζ_{it} is J -dimensional, realized indirect utility, v_{it} can be written strictly as a function

⁹This condition of our model ensures that consumers never devote all of their available resources to expenditure, desiring instead to carry liquidity forward in time. Since the dynamics of the model are sequential rather than in terms of continuation values, if consumers care about balance-matching, as Gathergood et al. (2019) have shown, we expect that they also care about balance holdings and how their behavior contributes to such holdings. For other "money in the utility function" models readers should refer to the monetary economics literature (Brock, 1974; Calvo, 1983; Obstfeld and Rogoff, 1983; Feenstra, 1986; Barnett et al., 1992; Walsh, 2010).

¹⁰In a consumption/savings model with borrowing, such as those featured in Bewley (1986), Huggett (1993), Aiya-gari (1994), and Huggett (1996), m_i is a limit to draw upon all available credit. The interpretation of m_i in the context of our model is more narrow. Rather, m_i represents the amount a consumer could feasibly over-draft their account. The mechanism is only needed for the rare occasion that over-drafting is observed, and it is not a critical component of the model. Indeed, in practice m_i is never binding.

¹¹We present a proof of this in Lemma B.1 in Supplemental Appendix B.

of the expenditure shocks, prices, the budgeting parameters, period- t income, and balances:

$$v_{it}(\zeta_{it}; \mathbf{p}_t, \boldsymbol{\omega}_{it}, \ell_{it}, b_{it}) = \sum_{j=1}^J \alpha_{ij} \ln \left(\frac{\omega_{ijt} + \zeta_{ijt}}{p_{jt}} + 1 \right) + \alpha_{i,J+1} \ln \left(\ell_{it} - \sum_{j=1}^J [\omega_{ijt} + \zeta_{ijt}] + m_i + r_t b_{it} \right) \quad (3)$$

The sense in which the doer is myopic can be seen by noting that, given income, balances, budgets, and prices, realized utility depends entirely on the realization of ζ_{it} . Thus, from the perspective of the doer, utility is random.

But, what is ζ_{it} ? This shock encodes unanticipated deviations from the spending plan, due to everything from weather-related variation to unexpected health shocks (i.e., taste shocks) to unexpected price shocks. Let ζ_{it}^u be a J -dimensional vector of the components of ζ_{it} corresponding to the preference shocks, and let ζ_{it}^p be a J -dimensional vector of the components of ζ_{it} which correspond to the price shocks. We assume orthogonality conditions such that $\zeta_{ijt} \perp \zeta_{ij',t}$ and $\zeta_{ijt}^u \perp \zeta_{ijt}^p$ where $j' \neq j$. This means that across categories, expenditure shocks are uncorrelated, and within a particular category, preference shocks are uncorrelated with price shocks.

Finally, set $\zeta_{it} = \zeta_{it}^u + \zeta_{it}^p$ and let the j^{th} component of ζ_{it} be mean-zero normally distributed — $\zeta_{ijt} \sim_{iid} \mathcal{N}(0, \sigma_{ij}^2)$. Note that when it comes to our quantitative strategy, we seek to infer budgeting behavior from expenditure data, but we do not observe prices or quantities. It is thus not possible to separately identify the degree to which the shock, ζ_{it} , results from preference heterogeneity versus unexpected price variation. For validation of our theory, though, this does not matter, because it does not matter *how* consumer budgets are violated (e.g., from preference shocks versus price shocks) but rather how they *respond* to such violations by updating their budgets.

3.2 The Planner

While the doer takes budgets, $\boldsymbol{\omega}_{it}$, as given, the planner must first decide what these budgets shall be conditional upon various behavioral and cognitive processing constraints. We now outline the planner's ex-ante decision process around which the doer's myopic expenditure decisions vary.

Mental Accounting. We start our description of the budgeting problem faced by the planner by first describing how the mental accounting state variable evolves. Denote the consumer's "mental

account” by a_{ijt} . This is a state variable that encodes the amount the consumer over- or under-spent on category j in the previous period, $t - 1$, relative to their budget. We loosely follow the mental accounting model in [Thaler \(1985\)](#), where past over- or under-expenditure in a particular category affects an individual’s willingness to engage in future expenditure. The law of motion for mental account balances is

$$a_{ijt} = \omega_{ij,t-1} - x_{ij,t-1} = -\zeta_{ij,t-1} \quad (4)$$

The mental account balance for consumption category j is exactly equal to the negative of that category’s idiosyncratic expenditure shock in the previous period. If $a_{ijt} < 0$ then the consumer over-spent in period $t - 1$, while if $a_{ijt} > 0$ they under-spent relative to their budget. The vector $\mathbf{a}_{it}$ is $J + 1$ dimensional where the $J + 1$ component, $a_{i,J+1,t}$, encodes how much a consumer over- or under-saved relative to their budget. Note that if $a_{i,J+1,t} < 0$ then the consumer has under-spent (over-saved) relative to their budget and vice-versa for $a_{i,J+1,t+1} > 0$. We assume that $\sum_{j=1}^{J+1} a_{ijt} = 0$ in every period, so that consumers’ beliefs regarding leftover resources from the previous period do not shift their total resource constraint. That is, their perception of how much they over- or under-spent last period in various commodity groups is exact.¹²

Budgeting Function. We defined the category j budget in period t as ω_{ijt} . This object is a function of the consumer’s mental account balance carried forward from the previous period, as well as their income, according to:

$$\omega_{ijt} = \theta_{ijt}\ell_{it} + \gamma_i a_{ijt} \quad (5)$$

In (5) γ_i captures how much a consumer’s mental account balance influences their budget. γ_i is restricted to reside in the unit interval. As $\gamma_i \rightarrow 1$ the degree to which a consumer anchors their spending to the amount over- or under-spent last period increases. At $\gamma_i = 0$ there is no mental-accounting anchoring effect.

Assuming γ_i is a fixed, agent-specific parameter and that consumers take their income flows as given, then in (5) the only object that consumers can ex-ante control is θ_{ijt} , which is the quasi-share

¹²The mental accounting model in [Farhi and Gabaix \(2020\)](#) features the same implicit assumption.

of income, ℓ_{it} , the consumer intends to spend on category j consumption in period t . This object is a quasi-share because it need not be interior to the unit interval: for example, if a consumer enters period t with a large mental account balance, a_{ijt} , due to under-spending in the previous period, the consumer may choose to set $\theta_{ijt} < 0$, thus reducing their period- t budget, ω_{ijt} , such that it is less than the effective mental account balance (i.e., $\omega_{ijt} < \gamma_i a_{ijt}$). Indeed, we assume that when making optimal budgeting decisions, the planner affects ω_{ijt} by setting θ_{ijt} , given values for ℓ_{it} and a_{ijt} which they know at the beginning of the period.

Cognitive Constraints on Budget Changes. While consumers choose θ_{ijt} when making an optimal budgeting decision, they do not necessarily make such decisions for every consumption category j in every period t because they face cognitive processing constraints. Let ψ_{ij} be the probability a consumer re-evaluates their budget-weighting variable θ_{ijt} in the first stage of period t . Let Γ_{ijt} be an indicator variable that equals 1 when a re-evaluation is made for the budget in j and 0 otherwise. Γ_{ijt} is Bernoulli distributed

$$\Gamma_{ijt} \underset{iid}{\sim} \text{Bernoulli}(\psi_{ij}) \quad (6)$$

with Γ_{ijt} assumed to be orthogonal to $\Gamma_{i'j',t}$ for all $j' \neq j$. This independence assumption induces narrow choice bracketing and results in a sparse-max budgeting structure, like the decision process analyzed in [Gabaix \(2014\)](#). In the context of our problem, a consumer choosing how to allocate funds across multiple different consumption budgets might only optimally re-evaluate one or two of those budgets in any given period, leaving the remainder fixed. In this way $\underline{k}_{it} = \sum_{j=1}^J \Gamma_{ijt}$. Note that just because a budget was re-evaluated does not necessarily mean that the new, candidate budget will be accepted by the consumer. This is because the absolute difference between the new candidate budget and the old budget must exceed some exogenously imposed numeracy threshold, which we discuss more below.

That in some periods $\underline{k}_{it} < J$ (strictly) is a modeling choice which draws on psychological evidence that consumers are cognitively constrained with regards to the amount of information they can process.¹³ Constrained consumers may possibly re-evaluate and subsequently update only a subset, $0 \leq \underline{k}_{it} \leq J$, of their budgets in any given period, where $\underline{k}_{it}$ describes the total

¹³This is synonymous with the number of choices they can consider at one time ([Miller, 1956](#); [Simon, 1957](#); [Cowan, 2000](#)).

number of expenditure budgets the agent optimally adjusts in a period.^{14,15} This integer constraint is exogenous and probabilistic: our premise is that consumers do not choose how rational they want to be.

Information Set. At the beginning of every period the planner realizes the vector of budget-evaluation indicators, $\underline{\Gamma}_{it}$. Consumers remain uncertain about ζ_{it} until the second stage of the period when the doer is in action. The planner also enters each period knowing the vector of budget weights, $\theta_{i,t-1}$, ascribed to last period’s income and the mental account balances, a_{it} . From the consumer’s perspective $\{\theta_{i,t-1}, a_{it}\}$ comprises the entirety of the set of period- t endogenous state variables. Additionally, consumers know the following exogenous state variables prior to setting their expenditure budgets — income ℓ_{it} , bank balances b_{it} , rates of return on such balances r_t , and the vector of average commodity-group aggregate price levels $\bar{p}_t$. Note that $\bar{p}_t$ is different from the actual price the consumer faces at the check-out counter. Rather, this price level can be thought of as an index, like the Consumer Price Index (CPI) or Personal Consumption Expenditure (PCE) index, which encode the relative value of baskets of various goods.

Numeracy Constraints. Prior to characterizing the optimality conditions associated with the expenditure budgets, we briefly describe how the numeracy constraints work, though this will become more obvious upon reading ahead. First, in any period and for any consumption category for which $\underline{\Gamma}_{ijt} = 1$, the consumer will choose a new candidate budget so as to maximize utility. This candidate consumption-category j budget will only be accepted as a new, optimal budget, if it is *significantly different enough* (i.e., satisfies the numeracy constraint) from the budget entering the period, $\theta_{ij,t-1}$. If the planner rejects a budget because it does not satisfy the numeracy condition, then we treat such a condition as the planner having not changed the budget at all, thus setting $\theta_{ijt} = \theta_{ij,t-1}$. We introduce the new state variable Γ_{ijt} (sans underline) which is equal to unity *if and only if* both $\underline{\Gamma}_{ijt} = 1$ and the candidate optimal budget satisfies the numeracy constraint. Thus,

¹⁴In some periods they may update the full set of budgets, and in some periods they may update none of their budgets.

¹⁵Imposing this integer constraint on the number of changes that can be made, rather than the magnitudes of the budget updates themselves, is supported by findings in Leslie et al. (2008) who argue that integer representations are innate within individual cognitive processes. For example, findings in Miller (1956) and Simon (1957) suggest that individuals can consider at most seven “chunks” of information at once. Meanwhile, Cowan (2000) says this number is closer to four. Further, information processing capacity in human memory is fairly limited and has been shown to impact consumption decisions (Malhotra, 1982). There is also a relationship between our work and recent work showing that agents exhibit aversion to complex decision making and prefer decisions wrought from more simplified rules (Oprea, 2020).

it must always be the case that $\Gamma_{ijt} \leq \underline{\Gamma}_{ijt}$. Further, letting $k_{ijt} = \sum_{j=1}^J \Gamma_{ijt}$ count the number of actually implemented budget changes which occur, it is necessarily so that $k_{ijt} \leq \underline{k}_{ijt}$ always, with equality only happening if all proposed budgets satisfy the numeracy constraints.

3.3 Choosing Expenditure Budgets

In each period prior to realizing ζ_{it} , the planner must set expenditure budgets subject to the constraints imposed by the behavioral frictions. Meanwhile, the doer's decision is exactly determined by the expenditure constraint in (2).¹⁶

Optimal Candidate Budgets. Recall that the vector θ_{it} is J -dimensional. Denote ϑ_{it} as the $\underline{k}_{it}$ -dimensional vector which holds, in cardinal order, the adjustable budgeting parameters of all commodities j for which $\underline{\Gamma}_{ijt} = 1$. These are the new, candidate budgets that must be evaluated against the numeracy constraint.¹⁷ Let $\mathbb{E}_{it}^{\zeta}$ be the expectation operator over the random variable ζ_{it} . After substituting (5) into (3) and replacing p_{jt} with the price index $\bar{p}_{jt}$, we can write expected indirect utility as

$$\begin{aligned} \mathbb{E}_{it}^{\zeta} v_{it}(\theta_{it}; \zeta_{it}) = \mathbb{E}_{it} \left\{ \sum_{j=1}^J \alpha_{ij} \ln \left(\frac{\theta_{ijt} \ell_{it} + \gamma_i a_{ijt} + \zeta_{ijt}}{\bar{p}_{jt}} + 1 \right) \right. \\ \left. + \alpha_{i,J+1} \ln \left(\ell_{it} - \sum_{j=1}^J [\theta_{ijt} \ell_{it} + \gamma_i a_{ijt} + \zeta_{ijt}] + m_i + r_t b_{it} \right) \right\} \end{aligned} \quad (7)$$

In (7) we suppress the explicit listing of arguments for known (from the planner's perspective) state variables in the definition of $v_{it}(\cdot)$. Note that (7) is the objective function the planner seeks to evaluate, subject to the integer constraints imposed by $\underline{\Gamma}_{it}$.

Let $\iota_{it}(\underline{\Gamma}_{it})$ be a vector-valued integer mapping that maps the index of components of ϑ_{it} back into the index of components of θ_{it} . This mapping is $\iota_{it}(\underline{\Gamma}_{it}) : \mathbb{N}^J \rightarrow \mathbb{N}^{\underline{k}_{it}}$ and valid for $\underline{k}_{it} > 0$. Let ϑ_{it}^* be the optimally-chosen analog of ϑ_{it} , prior to numeracy-constraint evaluation. The mapping $\iota_{it}(\underline{\Gamma}_{it})$ outputs in cardinal order the index of the components of θ_{it} to which we can assign the components of ϑ_{it}^* .¹⁸ Let y index the components of $\iota_{it}(\underline{\Gamma}_{it})$, so that ι_{iyt} denotes the y^{th} component

¹⁶In this sense our formulation aligns with the theory of [Ilut and Valchev \(2023\)](#) that decisions can be thought of as either "automatic" or "analytical." Here, the planner is the analytical/boundedly rational thinker, while spending is something that just happens automatically, though still constrained by the planner's budget and the distribution of the shock process.

¹⁷Note that ϑ_{it} is a sub-vector of θ_{it} corresponding to the non-zero indices of $\underline{\Gamma}_{it}$.

¹⁸For example, suppose $J = 4$ and $\underline{\Gamma}_{it} = (0, 1, 0, 1)$. Then this implies $\underline{k}_{it} = 2$ and $\iota_{it}(\underline{\Gamma}_{it}) = (2, 4)$.

of the vector $\boldsymbol{\iota}_{it}$. Finally, let $\boldsymbol{\theta}_{it}^*$ be the planner's optimally-chosen set of *candidate* budget shares (prior to numeracy-constraint evaluation), which also includes budgets in categories for which $\underline{\Gamma}_{ijt} = 0$. Then, these objects are

$$\begin{aligned} \boldsymbol{\vartheta}_{it}^* &= \underset{\boldsymbol{\vartheta}_{it}}{\operatorname{argmax}} \mathbb{E}_{it}^{\zeta} v_{it}(\boldsymbol{\vartheta}_{it}; \boldsymbol{\theta}_{it} \setminus \boldsymbol{\vartheta}_{it}, \boldsymbol{\zeta}_{it}) \\ \text{subject to } \boldsymbol{\theta}_{ijt}^* &= \begin{cases} \theta_{ij,t-1} & \underline{\Gamma}_{ijt} = 0 \\ \vartheta_{iyt}^* & \underline{\Gamma}_{ijt} = 1 \text{ \& } j = \iota_{iyt} \end{cases} \end{aligned} \quad (8)$$

In (8) the function $v_{it}(\boldsymbol{\vartheta}_{it}; \boldsymbol{\theta}_{it} \setminus \boldsymbol{\vartheta}_{it}, \boldsymbol{\zeta}_{it})$ has been re-written so that the set of elements of $\boldsymbol{\theta}_{it}$ for which $\underline{\Gamma}_{ijt} = 0$ is $\boldsymbol{\theta}_{it} \setminus \boldsymbol{\vartheta}_{it}$ and is taken as given by the planner (i.e., these budgets are not adjusted as part of the optimization problem). The solution to (8) is unique for large enough values of $\alpha_{i,J+1} > 0$, following from Lemma B.1 in Supplemental Appendix B and the fact that q_{ijt} is strictly increasing in θ_{ijt} .¹⁹ Clearly, if $\underline{k}_{it} = 0$ because $\underline{\Gamma}_{ijt} = 0, \forall j$, then we set $\boldsymbol{\theta}_{it} = \boldsymbol{\theta}_{i,t-1}$, and the budget shares do not change. However, note that as long as $\boldsymbol{a}_{it} \neq \boldsymbol{a}_{i,t-1}$, the total budget $\boldsymbol{\omega}_{it}$ (i.e., ex-ante expected expenditure under the assumption that $\boldsymbol{\zeta}_{it}$ is mean zero) *will* change (i.e., $\boldsymbol{\omega}_{it} \neq \boldsymbol{\omega}_{i,t-1}$), since $\boldsymbol{\omega}_{it}$ depends on $\boldsymbol{a}_{it}$.

For $\underline{k}_{it} > 0$ a candidate optimal choice of ϑ_{iyt}^* must satisfy

$$\mathbb{E}_{it}^{\zeta} \left\{ \frac{\partial v_{it}}{\partial \vartheta_{iyt}} \right\} = \mathbb{E}_{it}^{\zeta} \left\{ \frac{\partial u_i}{\partial q_{ijt}} \frac{\partial q_{ijt}}{\partial \vartheta_{iyt}} + \frac{\partial u_i}{\partial \bar{z}_{it}} \frac{\partial \bar{z}_{it}}{\partial \vartheta_{iyt}} \right\} = 0, \quad \forall y > 0 \text{ and } j = \iota_{iyt}, \quad (9)$$

which is equivalent to the problem's representation in (8). Due to the money-in-the-utility-function structure, any allocation that solves the consumer's problem will ensure (9) is satisfied, as consumers equate the expected marginal utility of additional consumption with the expected marginal utility of additional liquidity. Further, note that (9) is an equivalent, component-wise representation of the optimization problem specified (8): when partitioning $\boldsymbol{\theta}_{it}$ into $\boldsymbol{\vartheta}_{it}$ and $\boldsymbol{\theta}_{it} \setminus \boldsymbol{\vartheta}_{it}$, the optimization problem can be represented in an unconstrained form so that each component of the vector $\boldsymbol{\vartheta}_{it}^*$ must simply satisfy (9).

We can invert (9) under our utility parameterization to arrive at an analytical expression for

¹⁹Note that $\frac{\partial q_{ijt}}{\partial \theta_{ijt}} = \ell_{it}/p_{jt} > 0$, always.

ϑ_{it}^* . For some y indexing ι_{it} , let $\vartheta_{i,-y,t}^*$ denote the vector of optimally chosen budget candidates which does not include y . Note that this vector may be empty when no changes are made in a given period. The optimal budget share $\vartheta_{iyt}^*(\vartheta_{i,-y,t}^*)$ for good ι_{iyt} can be expressed as:

$$\begin{aligned} \vartheta_{iyt}^*(\vartheta_{i,-y,t}^*) = \mathbb{E}_{it} \left\{ \frac{\alpha_{i,\iota_{iyt}} \ell_{it} - \alpha_{i,\iota_{iyt}} \sum_{s \neq y} (\vartheta_{ist}^* \ell_{it} + \gamma_i a_{i,\iota_{ist},t} + \zeta_{i,\iota_{ist},t})}{\ell_{it} (\alpha_{i,\iota_{iyt}} + \alpha_{i,J+1})} \right. \\ - \frac{(\alpha_{i,\iota_{iyt}} + \alpha_{i,J+1}) (\gamma_i a_{i,\iota_{iyt},t} + \zeta_{i,\iota_{iyt},t}) + \alpha_{i,J+1} \bar{p}_{\iota_{iyt},t}}{\ell_{it} (\alpha_{i,\iota_{iyt}} + \alpha_{i,J+1})} \\ \left. - \frac{\alpha_{i,\iota_{iyt}} \left(\sum_{j \notin \iota_{it}(\underline{\Gamma}_{it})} (\theta_{ijt} \ell_{it} + \gamma_i a_{ijt} + \zeta_{ijt}) - m_i - r_t b_{it} \right)}{\ell_{it} (\alpha_{i,\iota_{iyt}} + \alpha_{i,J+1})} \right\} \end{aligned} \quad (10)$$

where $\theta_{ijt} = \theta_{ij,t-1}$ for $j \notin \iota_{it}(\underline{\Gamma}_{it})$.²⁰ Recall we assume that ℓ_{it} and $\bar{p}_t$ are ex-ante known. When estimating the structural model without explicitly observing budget-updating behavior, we exploit the independence of the components of ζ_{it} to iteratively sample the latent shocks $\underline{\Gamma}_{ijt}$ and update the components of ϑ_{it}^* accordingly, taking $\vartheta_{i,-y,t}^*$ as given. This allows for estimation of the mental accounting and budgeting parameters without having to iterate over the component functions of ϑ_{it} as we progress through a Markov Chain Monte Carlo estimation routine.

Numeracy Constraints. From our perspective as modelers, numeracy constraints must be evaluated *after* candidate budgets are computed, since one must have the candidate budget in hand to compare against the previous budget and then evaluate whether or not the numeracy constraint is satisfied. In terms of consumers' actual behaviors though, one can simply assume that the entire process characterized in §3.3 happens *all at once*: consumers solve (8) and evaluate the numeracy constraints we describe in this section simultaneously to arrive at period t 's optimal budget shares.

We incorporate numeracy constraints into the optimal budgeting problem for practical issues: if a budget change is really small, then it will have very little impact. We call the margin whereby a possible budget re-evaluation could occur (described above) the “extensive margin.” Let $\Gamma_{ijt} = 1$ (no underline) when both the extensive margin and the numeracy margin (intensive) for commodity group j are satisfied. Since re-evaluation occurs before the numeracy constraint is assessed, $\Gamma_{ijt} \leq \underline{\Gamma}_{ijt}$ always. Let k_{it} be the total number of budget updates inclusive of the

²⁰For the algebra behind this expression, see Supplemental Appendix A.

numeracy-evaluation condition, so that $k_{it} \leq \underline{k}_{it}$ always, as well.²¹ We consider two possible types of numeracy constraints, which are evaluated after the initial draws of $\underline{\Gamma}_{ijt}$:

- i. The candidate budget will be accepted and $\Gamma_{ijt} = 1$ if the constraint ϵ on the minimum percent-change of the budget in question is satisfied: $|\vartheta_{iyt}^* - \theta_{i,\iota_{iyt},t-1}| \geq \epsilon$.
- ii. The candidate budget will be accepted and $\Gamma_{ijt} = 1$ if the constraint ϵ_ℓ on the absolute dollar-value of the budget change in question is satisfied:²² $\ell_{it} |\vartheta_{iyt}^* - \theta_{i,\iota_{iyt},t-1}| \geq \epsilon_\ell$.

Note that both ϵ and ϵ_ℓ will be exogenously imposed by the econometrician.²³

In a slight abuse of notation, we will let $\iota_{it}(\Gamma_{it}) \subseteq \iota_{it}(\underline{\Gamma}_{it})$ be the possible subset of components of the extensive-margin index list for which the numeracy constraints are also satisfied.²⁴ From here on, assume any mention of ι_{it} without its dependency argument is such that $\iota_{it} \equiv \iota_{it}(\Gamma_{it})$. Thus, in our analyses we focus only on budget changes for which both the extensive margin and the numeracy constraints are met.

Note that, in practice, imposing the numeracy constraint amounts to assuming the consumer first receives a draw of $\underline{\Gamma}_{ijt} = 1$. Next, the consumer solves for the candidate budget ϑ_{iyt}^* in (10). Finally, if such a budget satisfies the numeracy constraint, then $\Gamma_{ijt} = \underline{\Gamma}_{ijt} = 1$, and the new budget is accepted. Otherwise $\Gamma_{ijt} = 0$ and $\theta_{ijt} = \theta_{ij,t-1}$.

3.4 Personal Mental Accounting Equilibrium

Figure 1 summarizes the flow of the decision process within a period, including the within-period timing of the different shock processes. First, the candidate re-evaluation vector $\underline{\Gamma}_{it}$ is realized. Next, optimal candidate budgets are computed if a re-evaluation is warranted. In the third and final step of the planner's decision process, the candidates are evaluated against the numeracy constraints; if no numeracy constraint is present, then we skip to the doer's problem. After the planner sets budgets, the vector of expenditure shocks, ζ_{it} , is realized $\forall j$. Period- t expenditure takes into account the mental account balance, a_{ijt} , which was set at the end of period $t - 1$. Next,

²¹It should be apparent that we use underlines to describe indicators for the extensive margin prior to the intensive-margin numeracy evaluation taking place.

²²The ℓ -subscript on ϵ_ℓ indicates that the constraint parameter has the same units as income (dollars).

²³Prior to estimation we must assume that either the consumer is constrained to make certain percentage-based updates or absolute dollar-based updates.

²⁴Note that $\iota_{it}(\Gamma_{it}) : \mathbb{N}^J \rightarrow \mathbb{N}^{k_{it}}$, instead of $\mathbb{N}^{k_{it}}$.

after all expenditure is realized, new mental account balances are set and then carried forward into $t + 1$. The process then repeats itself, starting with the realization of the budget re-evaluation shock in period $t + 1$.

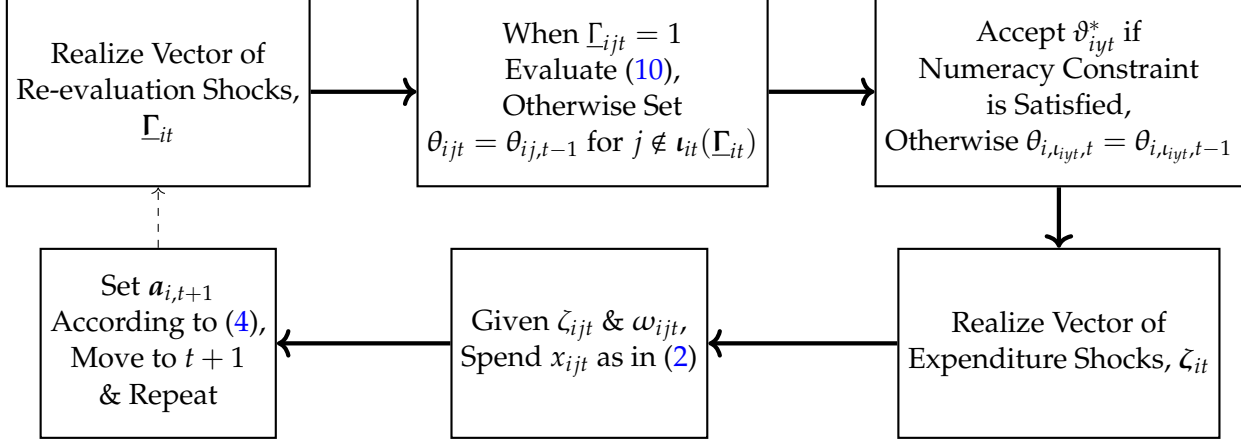


Figure 1: This directed graph presents the timing of the decision process within the period.

In Supplemental Appendix B we provide lemmas which characterize the conditions under which optimal budgets are unique and for which we can solve for such outcomes. Note that uniqueness holds almost always, as shown by Lemma B.2, so it does not threaten the validity of our estimation. We now formally define a “personal mental accounting equilibrium.”

Definition (Personal Mental Accounting Equilibrium): Assume Lemma B.2 holds. Let t_{i0} be the period in which an individual consumer enters the economy as an autonomous, decision-making agent. Given a numeracy threshold level (either ϵ or ϵ_ℓ), a sequence of prices $\{\bar{p}_t, r_t\}_{t \geq t_{i0}}$, a sequence of positive income values $\{\ell_{it}\}_{t \geq t_{i0}}$, a sequence of idiosyncratic expenditure shocks $\{\zeta_{it}\}_{t \geq t_{i0}}$, a sequence of idiosyncratic cognitive shocks $\{\underline{\Gamma}_{it}\}_{t \geq t_{i0}}$, and initial values for the budget weights, mental account balances, and bank balances $\{\theta_{i,t_{i0}}, \mathbf{a}_{i,t_{i0}}, b_{i,t_{i0}}\}$, a personal mental accounting equilibrium consists of:

- i. Sequences of policies: $\{q_{it}, z_{it}, \theta_{it}\}_{t \geq t_{i0}}$.
- ii. Sequences of balances: $\{\mathbf{a}_{it}, b_{it}\}_{t \geq t_{i0}}$.

such that in each period

- a. Given $\underline{\Gamma}_{it}$, $\mathbf{a}_{it}$, and b_{it} , θ_{it} satisfies the sparse-max indirect-utility maximization program in (8) and corresponding numeracy constraints.

- b. Given θ_{it} , ℓ_{it} , and $\mathbf{a}_{it}$, ω_{it} is exactly (5).
- c. Given θ_{it} , ℓ_{it} , ω_{it} , and ζ_{it} , $\mathbf{x}_{it}$ satisfies the expenditure criterion in (2).
- d. Given $\mathbf{x}_{it}$ and ℓ_{it} , $\mathbf{z}_{it}$ satisfies the marginal savings rule.
- e. Mental account balances, $\mathbf{a}_{it}$, are updated according to (4) with $a_{i,J+1,t} = -\sum_{j=1}^J a_{ijt}$.
- f. Bank balances evolve according to $b_{i,t+1} = r_t b_{it} z_{it}$.

4 Weekly Expenditure Data

To test the applicability of various behavioral theories to observed expenditure outcomes, we turn to a novel dataset of individual, consumer-level weekly-aggregated expenditure. The goal of our endeavor is to estimate consumer demand parameters while simultaneously considering the endogenous link between spending and budgeting decisions. Our challenge is that we only observe expenditure variation, not budgets.²⁵ Given model structure, we thus must latently infer how budgets and mental accounts evolve in order to quantitatively assess the degree to which expenditure data suggest consumers are rationally bounded.

Our dataset is an anonymized sample of low-income, pre-paid debit-card users from a large North American bank. We observe and classify users by card number, though multiple card numbers may be associated with a single household. Our most basic unit of observation is a “swipe” (i.e., card utilization) and/or a “load” (i.e., when income is added to a card, either by an employer’s direct deposit or some other kind of bank transfer). From swipes and loads, we sum up expenditure by category, income, and running balances to get weekly aggregates.

Expenditure is categorized by 4-digit Visa merchant category classification codes (MCC).²⁶ To classify expenditure categories we consider only the first two digits of the MCCs, which give us

²⁵An ideal dataset that tracks spending, prices, and income, along with consumers’ budgets, does not actually exist. Financial-planning and budgeting apps (e.g., Mint, Empower, YNAB (You Need a Budget), EveryDollar, Honeydue, and Personal Capital) record budgeting goals, sometimes by category, but such apps still do not record transaction-level prices. This means that categorical aggregation of expenditure would still need to happen in order to match spending to aggregated price indices. Further, users opt in to using such apps, meaning that conclusions drawn from their data are likely plagued by selection bias. Our expenditure data comes from enterprise banking data, which is not affected by such a bias. We can thus infer budgeting behavior from all types of consumers, including those who otherwise would not otherwise choose to use expenditure monitoring services.

²⁶Within the card industry this code is common across all card processors and is also known as the Standard Industry Code (SIC).

three explicit consumption categories and one commodity group that collects all other expenditure.²⁷ The explicit categories we examine, and their 2-digit MCCs are Groceries (54), Gasoline (55), and Food Away from Home (58).²⁸ The timeline of observations spans from September 2013 to January 2016 with most agents appearing as active users over only the latter part of this time period.

After data cleaning and selection, we have a total of $I = 2,509$ customers with a total of 71,859 weekly observations. This gives an average of approximately 29 weeks per customer. Importantly, as we alluded to above, we restrict our sample of consumers to those who appear to use their pre-paid debit card as their only banking product. These consumers are associated with deposits of regular, weekly income to their pre-paid card and use the card for expenditures on a regular basis. On one hand, restricting the analysis to pre-paid debit card users limits inferences to only a small subset of the population, namely the underbanked with limited access to other banking products, such as checking accounts and credit cards.²⁹ On the other hand given data limitations, doing so greatly increases the probability that the income and expenditure profiles we observe represent complete consumption profiles for those agents who use this particular product. Each consumer profile contains at least 16 consecutive weeks of regular income. This ensures that we are observing a regular income process for the consumer. Finally, we exclude from our sample consumers for whom we observe at least four consecutive weeks of zero expenditure in one or more of the aforementioned categories.

Summary statistics for our entire sample are presented in Table 1, where the units are agent-level averages.³⁰ The median consumer in our sample earns \$460.05 per week after taxes, which aggregates to \$23,922.60 per year in nominal dollars. For comparison, consider that median U.S. household income in 2015 was \$56,516 according to the U.S. Census bureau. 96.8% of our sample fall below 2015 median household income. The 2015 poverty thresholds for one, two, three, and

²⁷Note that the fourth category representing all remaining expenditure includes both classifiable transactions that fall outside the 2-digit codes above and cash withdrawals from the account. This means that some possibly classifiable expenditure may be mis-categorized. Unfortunately, this cannot be reconciled without making some very strong assumptions with regards to how consumers spend their money.

²⁸Our goal with category selection is to look at purchasing behavior in categories whose demand has also been widely analyzed in the literature. For the current analysis categories have been exogenously defined in order to discipline inference.

²⁹We have verified that these customers do not have other accounts at the bank from which our data originates. Although it is possible that these individuals have accounts at other banks that we cannot see, we believe this to be unlikely given the nature of the selected customer base.

³⁰For example, in Table 1 “Median” income corresponds to the median agent-level average income and so on.

Table 1: Summary Statistics Over Agent-level Means

	5%	1st Qu.	Median	Mean	3rd Qu.	95%	S.D.
Weeks	16	20	25	28.640	35	52	11.131
Income	207.096	338.855	460.052	510.732	620.131	981.994	258.288
Balances	6.216	89.953	204.734	500.744	438.464	1,415.661	1,868.651
<i>Expenditure</i>							
Groceries	11.472	22.282	34.715	42.018	54.195	94.548	30.622
Auto/Gas	13.393	29.191	48.430	60.941	80.198	148.186	46.064
Food Away	14.614	30.179	51.965	68.724	86.684	177.296	61.656
Other	99.211	194.847	283.406	326.124	415.037	687.897	195.551

Number of Consumers $I = 2,509$; Total Consumer Weeks $\sum_i \sum_{t_i} = 71,859$

four person households where the head was under the age of 65 were \$12,331, \$15,952, \$18,871, and \$24,257; 7.8%, 18.3%, 30.8%, and 51.4% of our sample respectively fall below these corresponding poverty thresholds.³¹

We do not observe purchase prices at the transaction-level, so we turn to the Consumer Price Index (CPI) for price indices. Price units are \$1982-84 dollars. All price indices are published at the monthly level, so we use simple linear interpolation to get weekly, aggregate price estimates. For the prices of groceries and food away from home, we use the U.S. city average food and beverage price index (series I.D. "CUUR0000SAF"). For the price of gasoline and other miscellaneous automotive maintenance expenses, we use the U.S. city average transportation price index (series I.D. "CUUR0000SAT"). All other expenditure will be deflated using the U.S. city average all items index (series I.D. "CUUR0000SA0").

5 Structural Model Estimation

Few economics papers that grapple with behavioral phenomena in a decision-theoretic structural manner attempt to estimate such models using non-experimental field data. A notable exception is [Hastings and Shapiro \(2018\)](#) who take a non-parametric approach to identifying and estimating the degree to which low-income consumers appear to be constrained by mental accounting behavior. Given one aim of this paper is to infer budgeting decisions without observing actual budgets, only expenditure, we desire a parametric model to avoid placing further burdens on the data. We

³¹We do not observe the household size in our dataset.

have so far described such a model that places structure on the agents' decisions. We can then exploit our assumed structure to estimate equilibrium budgeting decisions as parameters. We do this by turning to the literature in Bayesian statistics that deals with detecting latent change points of observed time series. Since our time series of budgeting decisions, ϑ_{it}^* , are themselves endogenous and unobserved, we require that the equilibrium condition in (10) and the equilibrium laws of motion of mental account balances in (4) all exactly hold conditional upon the time-independent preference parameters as well as estimated latent change points, which are identical to the vector of unobserved budget-updating indicators, Γ_{it} .³² This is accomplished with an algorithm that borrows from the endogenous change-point inference techniques described in McCulloch and Tsay (1993), Chib (1998), and Koop and Potter (2007) nested within a hierarchical Metropolis-Hastings (MH) within Gibbs Markov Chain Monte Carlo (MCMC) sampling algorithm (Hastings, 1970; Geman and Geman, 1984).

The key challenge of our structural estimation procedure is that we must estimate budget weights, mental accounts, and cognitive frictions as latent time series variables, $\{\theta_{it}, a_{it}, \Gamma_{it}\}_{t=1}^{T_i}$ for each i using only variation in observed expenditure. Due to the vast degree of observed heterogeneity across agents, most of our estimation routine operates only on agent-level parameters. Because of the need to sample latent time series of consumer decisions, the parameter space is large, containing over four million parameters.³³ Changes to budgets are conditionally identified by level shifts in average spending. The unknown initial mental account balance a_{i1} is conditionally identified by the first-period deviation from average spending. The budgets themselves are identified conditional upon the other parameters and the parametric structure of the model itself.

Due to its complexity of detail, we relegate a detailed description of the estimator, including the MH within Gibbs MCMC routine, to Supplemental Appendix D.1. Additional MCMC diagnostics are presented in Supplemental Appendix D.3. In this section we first discuss model fitness of our multiple estimators. We then quantitatively explore the implications of our estimates for behavioral features such as budgeting, anchoring, and choice bracketing.

³²We count budget updates only if they satisfy both the extensive and intensive (numeracy) margins.

³³The exact dimension of the parameter space in our preferred model is 4,664,212 with 4,598,976 of these parameters being the values of latent, time-dependent budgeting and mental accounting variables.

5.1 Model Fitness

We estimate several different model specifications using the MCMC algorithm. We then assess and compare the model fit of each in order to understand the degree to which assumptions regarding numeracy, budgeting frictions, and mental accounting affect predictive inference. With respect to numeracy, we consider six different thresholds, depending on whether we assume numeracy is constrained along the relative (percentage) or absolute (dollar) margins.³⁴

- i. Along the relative margin, we estimate four models with thresholds $\epsilon \in \{0, 0.01, 0.05, 0.1\}$.
- ii. Along the absolute margin, we estimate two models with thresholds $\epsilon_\ell \in \{0.01, 1\}$.

In the baseline model where $\epsilon = \epsilon_\ell = 0$, we count estimated budget updates anytime the candidate is such that $\vartheta_{iyt}^* \neq \theta_{i,t_{iyt},t-1}$. Here, in the main text we present estimation results from both the baseline and \$1-threshold models, which is the best-fitting model according to the information criterion we employ.³⁵

In addition to estimating six versions of the full model with different assumptions regarding numeracy, we also estimate several different models where different behavioral features are turned off. Specifically, we estimate the baseline model ($\epsilon = \epsilon_\ell = 0$) and the \$1-threshold model ($\epsilon_\ell = 1$) under the following separate assumptions:

- i. Heterogeneous rationality $\psi_{ij} \in (0, 1)$ and full anchoring, $\gamma_i = 1$.
- ii. No anchoring, $\psi_{ij} \in (0, 1)$ and $\gamma_i = 0$.
- iii. Constant budget weights, $\Gamma_{ijt} = 0$ and $\gamma_i = 1$.
- iv. Constant budget weights with no anchoring, $\Gamma_{ijt} = 0$ and $\gamma_i = 0$.
- v. No budget-updating frictions, $\Gamma_{ijt} = 1$ and $\gamma_i = 1$.
- vi. Initial mental account balances are always zero, $a_{ij,t_{i0}} = 0$, where t_{i0} is the first observation in the sample for agent i .

³⁴Recall that, for example, $\epsilon = 0.01$ implies that optimal budget updates are only recorded as such if the absolute difference between current and candidate expenditure shares is $|\vartheta_{iyt}^* - \theta_{i,t_{iyt},t-1}| > \epsilon$. For simulations where budget updates are bounded along the absolute margin, budget updates are only recorded as such if the absolute difference is $\ell_{it} |\vartheta_{iyt}^* - \theta_{i,t_{iyt},t-1}| > \epsilon_\ell$.

³⁵Including the \$1-dollar threshold model amongst our primary results is also a nod to the literature suggesting consumers exhibit so-called “left digit biases” (Lacetera et al., 2012; Sokolova et al., 2020)

Model (v) is designed to best mimic a classical two-stage budgeting problem with ex-ante preference uncertainty and constant expenditure share preference weights. Alternative models (iii) and (iv) are identical across numeracy constraints since we are assuming both the extensive and intensive optimization margins are non-binding always (i.e., both $\Gamma_{ijt} = 0$ and $\underline{\Gamma}_{ijt} = 0$). Model (vi) is included to assess how much our prior distributional assumptions for the initial mental account state variable drive the results (they do not).

In Supplemental Appendix D.2 we describe fitness summary statistics and present a table that compares all of the models by numeracy threshold and the various assumptions on their behavioral parameters. We use an information criterion, comparing predictive densities across models a la [Gelman et al. \(2013\)](#), to assess model fitness. Note that all fully-parameterized models ($\psi_{ij} \in (0, 1)$ and $\gamma_i = 1$) with non-zero numeracy constraints are marginally better fits than the baseline model. Empirically-speaking, the fully-parameterized \$1-threshold model achieves the largest information gain on the baseline model relative to any other fully-parameterized models. This empirical result conforms with theory and extensive evidence that individuals are often constrained by integer representations when making numerical decisions ([Miller, 1956](#); [Simon, 1957](#); [Cowan, 2000](#); [Leslie et al., 2008](#)). Given these fitness results, we focus most of our discussion on results generated by two models: 1) the baseline model with no numeracy constraints and 2) the \$1-threshold model, each with all of the mental accounting features (i.e., $\psi_{ij} \in (0, 1)$ and $\gamma_i = 1$). Both models allow for anchoring and heterogeneous rationality.

To visually illustrate how well both the baseline model and \$1-threshold model fit the data, Figure 2 presents the time series of expenditures (black lines) along with predictive means (red and blue lines) from the two fully-parameterized estimations with $\psi_{ij} \in (0, 1)$ and $\gamma_i = 1$ for a selected consumer whose average weekly income is closest to the sample median. Mean weekly expenditure in each category is as follows: groceries \$60.67, auto/gasoline \$32.50, food away from home \$21.98, and other expenditure \$357.00. This consumer earns an average of \$460.05 per week and maintains an average weekly card balance of \$115.21 after spending. Notice that weekly expenditure patterns are spiky yet weakly persistent — a pattern we observe for many consumers in our sample. Consequently, explaining these patterns requires a demand model, formalized by theory, with features that can endogenously explain spending spikes, persistence, and trend decline or growth in individual consumption expenditure series. The model we have developed

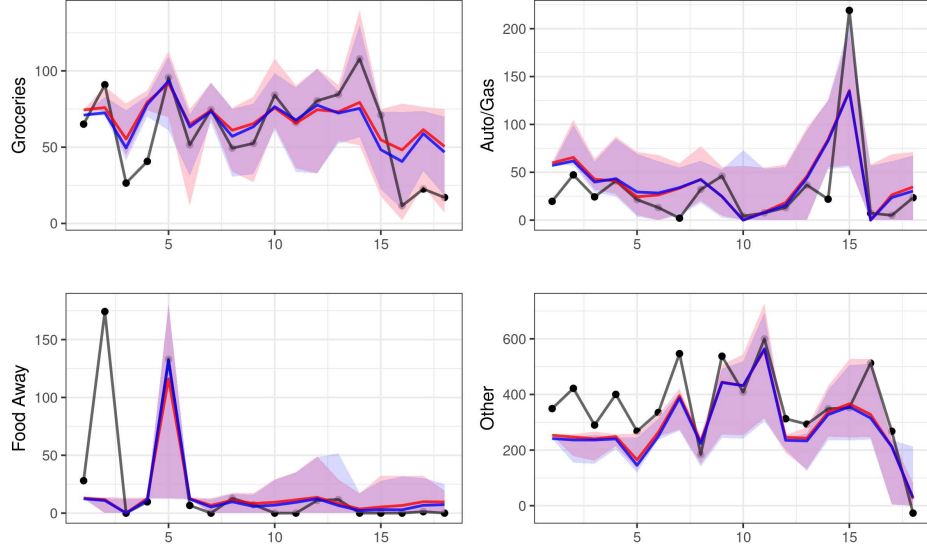


Figure 2: Black lines represent actual weekly spending for the median income agent. Red lines represent the baseline model’s predicted means with 95% confidence regions in pink. The blue lines represent the \$1-threshold model’s predicted means with 95% confidence regions in light blue. Notice that predictions from both models are almost identical.

can predict all of these features: spending spikes result from both endogenous mental accounting and exogenous taste shocks; persistence is a function of budgetary bounded rationality and mental accounting; and changes to average spending and trends are associated with budget updates.

Summary statistics over agent-level means for parameter estimates are presented in Table 2 for both the baseline model sans numeracy threshold and the \$1-threshold model.³⁶ Amongst the mental accounting parameters, the variances of the posterior distributions of agent-level mental account balance means, $\hat{a}_i$, are large. Some of the likelihood variance estimates, $\hat{\sigma}_{ij}^2$, are also large. These high values result from behavior whereby a consumer goes from very few (or no) purchases one week to high expenditure values the next. Such high values are not surprising given the spikiness of the various time series. Remember, $a_{ij,t+1} = -\zeta_{ijt}$, so high variance in ζ_{ijt} will lead to high variance in $\hat{a}_{ij}$ as well. This variation propagates through the model via optimal budget updates, ϑ_{iyt}^* . Lacking explicit budgeting data, this posterior variability is to be expected. Further, (though not presented here) we find that estimates at the fifth and ninety-fifth percentiles are reasonable for both models.

³⁶For parameters that are time dependent, we average both across MCMC draws and over time for each agent. For the global parameters, we simply take summary statistics over the outputted chain.

Table 2: Posterior Summary Statistics

Description	Parameter	Baseline Model $\epsilon = \epsilon_\ell = 0, \psi_{ij} \in (0, 1), \& \gamma_i = 1$		\$1-threshold Model $\epsilon_\ell = 1, \psi_{ij} \in (0, 1), \& \gamma_i = 1$	
		Mean	S.D. ^e	Mean	S.D. ^e
<i>Borrowing Limit</i>	m_i	5,311.618	2,686.200	5,311.618	2,686.200
<i>Long-run</i>	$\hat{\alpha}_{i1}^a$	0.106	0.064	0.106	0.064
<i>Expenditure</i>	$\hat{\alpha}_{i2}$	0.146	0.088	0.146	0.088
<i>Share</i>	$\hat{\alpha}_{i3}$	0.150	0.097	0.150	0.097
	$\hat{\alpha}_{i4}$	0.598	0.148	0.598	0.148
<i>Likelihood</i>	$\hat{\sigma}_{i1}$	29.346	24.325	34.604	200.396
<i>Standard</i>	$\hat{\sigma}_{i2}$	45.774	54.622	54.062	188.935
<i>Deviation</i>	$\hat{\sigma}_{i3}$	63.186	64.446	73.899	251.622
	$\hat{\sigma}_{i4}$	245.342	2,031.048	364.204	1,785.711
<i>Shape Parameter</i>	$\hat{\tau}_{i1}$	0.746	2.537	3.90×10^{10}	1.96×10^{13}
<i>for Likelihood</i>	$\hat{\tau}_{i2}$	0.978	3.305	2.18×10^{21}	1.09×10^{23}
<i>Variance</i>	$\hat{\tau}_{i3}$	0.903	3.625	0.353	1.997
<i>Prior</i>	$\hat{\tau}_{i4}$	0.566	2.009	1.29×10^{18}	6.44×10^{19}
<i>Scale Parameter</i>	$\hat{\beta}_{i1}$	1.737	2.501	4.657	167.197
<i>for Likelihood</i>	$\hat{\beta}_{i2}$	1.968	3.266	9.682	417.278
<i>Variance</i>	$\hat{\beta}_{i3}$	1.899	3.620	1.350	1.990
<i>Prior</i>	$\hat{\beta}_{i4}$	1.566	2.011	1,626.734	81,422.770
<i>Liquidity Preferences</i>	$\hat{\alpha}_{i,j+1}$	12.178	3.388	11.488	10.442
<i>Mean of $\hat{\alpha}_{i,j+1}$</i>	$\hat{\mu}_{\alpha_{i,j+1}}$	15.315	0.018	13.251	1.004
<i>Probability of</i>	$\hat{\psi}_{i1}$	0.667	0.194	0.545	0.207
<i>Budget</i>	$\hat{\psi}_{i2}$	0.644	0.197	0.545	0.201
<i>Update</i>	$\hat{\psi}_{i3}$	0.590	0.195	0.517	0.188
	$\hat{\psi}_{i4}$	0.536	0.231	0.515	0.227
<i>Budgeted Share</i>	$\hat{\theta}_{i1}^b$	0.204	2.005	0.114	0.108
<i>of Income</i>	$\hat{\theta}_{i2}$	0.330	4.495	0.156	0.146
<i>Chosen by</i>	$\hat{\theta}_{i3}$	0.451	6.954	0.171	0.168
<i>the Consumer</i>	$\hat{\theta}_{i4}$	2.188	41.125	0.708	0.295
<i>Cognitive and</i>	$\hat{\Gamma}_{i1}^c$	0.694	0.201	0.566	0.215
<i>Numeracy</i>	$\hat{\Gamma}_{i2}$	0.669	0.204	0.566	0.209
<i>Frictions w.r.t.</i>	$\hat{\Gamma}_{i3}$	0.613	0.202	0.537	0.195
<i>Budget Updates</i>	$\hat{\Gamma}_{i4}$	0.557	0.239	0.536	0.236
<i>Mental</i>	$\hat{a}_{i1}$	-0.182	35.937	3.663	305.530
<i>Account</i>	$\hat{a}_{i2}$	0.303	32.942	7.228	279.077
<i>Balances</i>	$\hat{a}_{i3}$	-2.942	65.058	6.223	372.457
	$\hat{a}_{i4}$	-158.191	8,548.589	211.587	2,594.666
<i>Standard</i>	$\hat{\sigma}_{a_{i1}}$	0.013	0.007	0.013	0.007
<i>Deviation</i>	$\hat{\sigma}_{a_{i2}}$	0.015	0.009	0.015	0.009
<i>for Mental</i>	$\hat{\sigma}_{a_{i3}}$	0.018	0.011	0.018	0.011
<i>Accounts</i>	$\hat{\sigma}_{a_{i4}}$	0.034	0.023	0.034	0.023
<i>Global Mean of $\hat{\mu}_{\alpha_{i,j+1}}$</i>	$\hat{\mu}_\mu^d$	14.844	3.395	13.074	7.117
<i>Global S.D. of $\hat{\mu}_{\alpha_{i,j+1}}$</i>	$\hat{\sigma}_\mu$	0.501	0.117	3.512	0.709

^a Second subscript corresponds to commodity group — groceries ($j = 1$), auto/gasoline ($j = 2$), food away from home ($j = 3$), and other expenditure ($j = 4$).

^b For time-dependent parameters, θ_{ijt} , Γ_{ijt} , and a_{ijt} , we average over both posterior draws and time for each agent and each commodity category. The statistics presented are summary statistics over these agent-level averages.

^c The agent-level means for the budget-updating indicators are taken from the \$1-threshold model where $\Gamma_{ijt} = 1$ if $\ell_{it} |\theta_{ijt} - \theta_{ij,t-1}| > \kappa_\ell = 1$.

^d $\hat{\mu}_\mu$ and $\hat{\sigma}_\mu$ are the only two global parameters. They are summary statistics over agent-level $\hat{\mu}_{\alpha_{i,j+1}}$. From simulations, we found that these parameters are needed to induce posterior shrinkage.

^e This is the standard deviation of the posterior distribution *not* the standard error of the posterior mean.

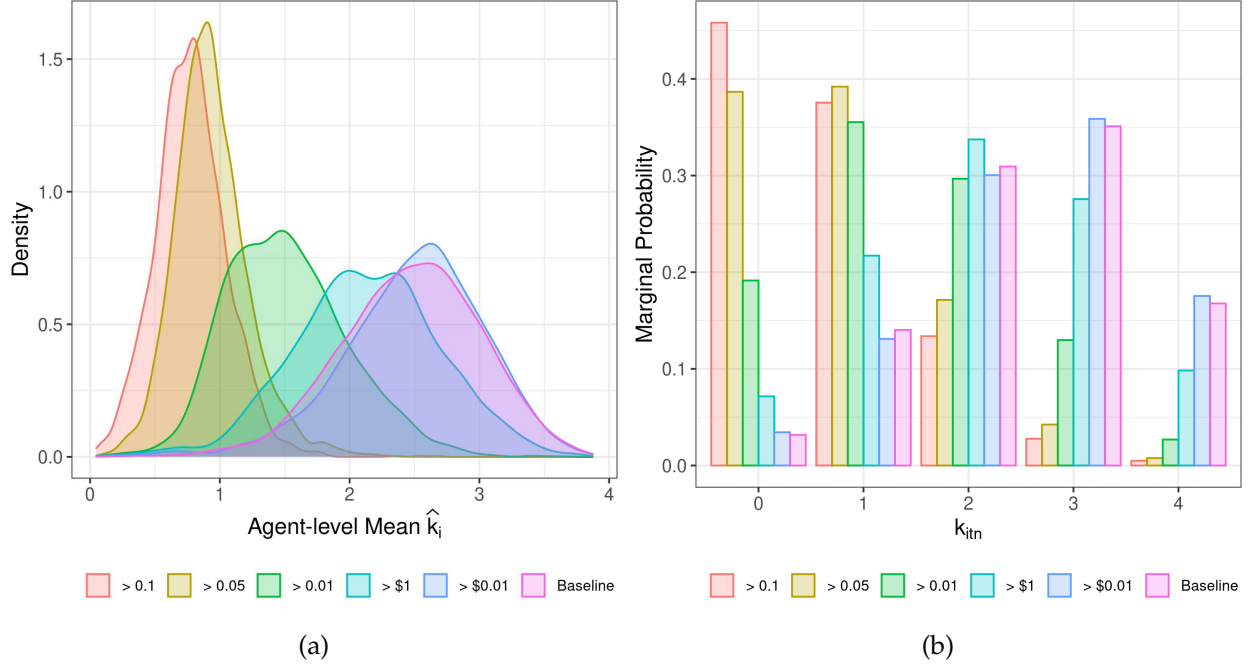


Figure 3: In panel (a) we present the posterior density of $\hat{k}_i = \frac{1}{NT_i} \sum_n \sum_t \sum_j \Gamma_{ijtn}$, i.e. the number of budget changes each period, which satisfy both attentiveness and numeracy constraints, averaged over time and across MCMC draws for each agent, using a smoothed Gaussian kernel with the Silverman rule-of-thumb for choosing the bandwidth (Silverman, 1986). In panel (b) we show the marginal posterior probability mass function for the atomic posterior draws (epochs) of k_{itn} . Colors denote different thresholds for judging when a budget update has occurred. In the baseline scenario we assume that any time $\theta_{ijtn} \neq \theta_{ij,t-1,n}$ a budget update has occurred, while the other scenarios correspond to those associated with numeracy thresholds $\epsilon \in \{0.1, 0.05, 0.01\}$ or $\epsilon_\ell \in \{1, 0.01\}$.

5.2 Budget-updating Behavior

Figure 3 provides visual summaries of the estimated total number of budget switches $k_{itn} = \sum_{j=1}^J \Gamma_{ijtn}$ under different numeracy thresholds. In the baseline model consumers make an average of 2.48 budget updates every week. When the absolute threshold is set at $\epsilon_\ell = 0.01$, so that we encode budget updates only when the absolute difference in old versus new budgets exceeds one cent, the average number of updates is 2.51 each week. Estimates of $\hat{k}_i$ between the baseline model and the model with $\epsilon_\ell = 0.01$ are not significantly (statistically) different from each other. When the \$1-threshold model is considered, we observe an average of 2.11 updates per week. These numbers fall when we consider different minimum relative thresholds — 1.45 when $\epsilon = 0.01$, 0.89 when $\epsilon = 0.05$, and 0.75 when $\epsilon = 0.1$. With a \$1 threshold (teal), full rationality ($k_{it} = J = 4$) is observed only 10% of the time. Even without allowing for numeracy constraints, just over 80% of consumer-week combinations exhibit some degree of bounded rationality providing strong ev-

idence for the presence of financial-attentiveness frictions.

Raising the numeracy threshold from zero to \$1 leads to approximately 0.34 fewer updates, on average, per week across our sample of consumers. When taking into consideration the relative thresholds, that value climbs to 0.95 fewer updates when $\epsilon = 0.01$, 1.54 fewer updates when $\epsilon = 0.05$, and 1.69 fewer updates when $\epsilon = 0.1$. Overall, total budget updates across consumers and over time drop 14.9% when the \$1 threshold is implemented. Under the relative thresholds, we observe a 41.8% drop when $\epsilon = 0.01$, a 64% drop when $\epsilon = 0.05$, and a 70% drop when $\epsilon = 0.1$.

Broadly, our results indicate that few consumers update all of their budgets every period, providing evidence for binding cognitive constraints and narrow bracketing. At the same time, these constraints do not appear to force agents to stick with budgets for long extended periods, since on average they still make a few updates every week. Thus, while agents do not consistently update their budgets every week, neither do they neglect these budgets over an extended amount of time.

5.3 Classifying Consumers By Budgetary Responsiveness

In this section we explore how expected total spending (or expected savings) and total realized spending (or realized savings) depend on total over- or under-expenditure from the previous period, $a_{i,J+1,t}$. Recall that over-saving in period $t - 1$ implies $a_{i,J+1,t} < 0$, not positive, so that over-spending in $t - 1$ is associated with $a_{i,J+1,t} > 0$. Consumers can respond to $a_{i,J+1,t}$ ex-ante by adjusting their budget shares and thus their expected *total* expenditure. While our modeling approach does not take an explicit stand on how consumers optimally choose their actual ex-post expenditure after having ex-ante set their optimal budgets, we can still examine the degree to which actual expenditure varies in the estimated state variable $a_{i,J+1,t}$.

We type-cast consumers based on the probability we observe a budget (ex-ante) or expenditure (ex-post) increase following over/under spending in the previous period (i.e., conditional upon the sign of $a_{i,J+1,t}$). Let $\hat{\mathbb{P}}_i(\cdot|\cdot)$ be a posterior-estimated conditional probability function. Let $\text{sgn}\{y\}$ be the sign function that describes whether or not the variable y is negative. The estimated probabilities we consider, depending on whether we are casting ex-ante or ex-post types,

are $\hat{\mathbb{P}}_i \left(\text{sgn} \left\{ \sum_j (\omega_{ijt} - \omega_{ij,t-1}) \right\} \middle| \text{sgn} \{a_{i,J+1,t}\} \right)$ and $\hat{\mathbb{P}}_i \left(\text{sgn} \left\{ \sum_j (x_{ijt} - x_{ij,t-1}) \right\} \middle| \text{sgn} \{a_{i,J+1,t}\} \right)$. We can classify consumers into four broad types based on their average behavior:^{37,38}

- i. (*Budget prioritizers* — negative correlation patterns) Those who respond to over-spending in $t - 1$ by reducing expected (or realized) spending in t and under-spending in $t - 1$ by increasing expected (or realized) spending in t . These consumers use mental accounting as a correction mechanism in the original spirit described in [Thaler \(1985\)](#), with the goal of balancing out prior over- or under-spending by spending less or more in the next period, so that their budgets remain fairly stable and disciplined, overall. These are consumers for whom the posterior observed probabilities are such that

$$\begin{aligned} \hat{\mathbb{P}}_i \left(\sum_j (\omega_{ijt} - \omega_{ij,t-1}) < 0 \middle| a_{i,J+1,t} > 0 \right) \\ = \frac{\sum_n \sum_t \mathbf{1} \{ \sum_j (\tilde{\omega}_{ijtn} - \tilde{\omega}_{ij,t-1,n}) < 0 \} \mathbf{1} \{ \tilde{a}_{i,J+1,tn} > 0 \} }{\sum_n \sum_t \mathbf{1} \{ \tilde{a}_{i,J+1,tn} > 0 \} } > 0.5 \end{aligned} \quad (11)$$

$$\begin{aligned} \text{and } \hat{\mathbb{P}}_i \left(\sum_j (\omega_{ijt} - \omega_{ij,t-1}) > 0 \middle| a_{i,J+1,t} < 0 \right) \\ = \frac{\sum_n \sum_t \mathbf{1} \{ \sum_j (\tilde{\omega}_{ijtn} - \tilde{\omega}_{ij,t-1,n}) > 0 \} \mathbf{1} \{ \tilde{a}_{i,J+1,tn} < 0 \} }{\sum_n \sum_t \mathbf{1} \{ \tilde{a}_{i,J+1,tn} < 0 \} } > 0.5 \end{aligned} \quad (12)$$

Together, the signs of (11) and (12) fully characterize a “budget prioritizer” (ex-ante type i). Replace $\tilde{\omega}$ ’s in (11) and (12) with actual x ’s (i.e., expenditure *data*) and you have the statistics that summarize a “spending prioritizer” (ex-post type i). Both (11) and (12) must hold for a consumer to be classified as a budget and/or spending prioritizer.

- ii. (*Spending followers* — positive correlation patterns) Those who respond to over-spending in $t - 1$ by increasing expected (or realized) spending in t and under-spending in $t - 1$ by reducing expected (or realized) spending in t . These consumers engage in persistent over-spending and under-spending over successive periods. When updating budgets, they often update them to be more in line with their prior spending, rather than the other way around. The

³⁷In what follows “realized” refers to ex-post types inferred from observed spending (not unobserved budgeting) behavior.

³⁸To some degree we engage in similar exercises to [Andre et al. \(2025\)](#) by type-casting consumers based on their behavior. Where our endeavors depart from theirs, however, is that they simply look at how people spend and save out of transitory income shocks, whereas we examine behavioral responsiveness to shocks conditional upon estimated, latent budgeting patterns.

ex-ante probabilities which must hold are $\hat{\mathbb{P}}_i\left(\sum_j(\omega_{ijt} - \omega_{ij,t-1}) > 0 \mid a_{i,J+1,t} > 0\right) > 0.5$ and $\hat{\mathbb{P}}_i\left(\sum_j(\omega_{ijt} - \omega_{ij,t-1}) < 0 \mid a_{i,J+1,t} < 0\right) > 0.5$ (and the same for ex-post).³⁹

- iii. (*Spendthrifts*) Those who respond to both over- and under-spending in $t - 1$ by increasing expected (or realized) spending in t . These consumers use under-spending as an excuse to spend more in the next period (i.e., use their surplus, implicit savings) and over-spending as a reason to increase their future budgets rather than decrease their spending. The ex-ante probabilities which must hold are $\hat{\mathbb{P}}_i\left(\sum_j(\omega_{ijt} - \omega_{ij,t-1}) > 0 \mid a_{i,J+1,t} > 0\right) > 0.5$ and $\hat{\mathbb{P}}_i\left(\sum_j(\omega_{ijt} - \omega_{ij,t-1}) > 0 \mid a_{i,J+1,t} < 0\right) > 0.5$ (and the same for ex-post).
- iv. (*Frugal savers*) Those who respond to both over- and under-spending in $t - 1$ by decreasing expected (or realized) spending in t . These consumers use under-spending as a motivator to save more (i.e., continue spending less) in the next period and respond to over-spending by reducing their future budget.⁴⁰ The ex-ante probabilities which must hold are $\hat{\mathbb{P}}_i\left(\sum_j(\omega_{ijt} - \omega_{ij,t-1}) < 0 \mid a_{i,J+1,t} > 0\right) > 0.5$ and $\hat{\mathbb{P}}_i\left(\sum_j(\omega_{ijt} - \omega_{ij,t-1}) < 0 \mid a_{i,J+1,t} < 0\right) > 0.5$ (and the same for ex-post).

We can also characterize the joint distribution of consumers across both how their ex-ante budgeting behavior and ex-post spending behavior relate to $a_{i,J+1,t}$. This more granular type casting yields $4^2 = 16$ different consumer types. In what follows we present the marginal distributions of consumers pertaining to their particular ex-ante and ex-post type, analyzing how income and savings vary across types to understand if any systematic relationships exists. Then, we repeat the exercise examining the joint distribution of ex-ante and ex-post types. Finally, we focus on the spending and savings patterns for the subset of consumers who are of different ex-ante and ex-post types in order to understand why, between budgeting and spending, type-inconsistent behavior occurs.

³⁹The empirical counterparts of all probabilities are computed in the manner as (11) and (12). We avoid the algebraic details here to save space.

⁴⁰Type (iii) and (iv) consumers fall on the so-called “spendthrift” and “tightwad” spectrum proposed in Rick et al. (2008). Spendthrifts lack in self-regulation, experiencing low average dis-utility with respect to over-spending, while tight wads are overly sensitive to over-spending.

Table 3: Marginal Distributions of Ex-ante and Ex-post Types ^a

<i>Expected Expenditure</i>				
Type				
Model ^b	(i)	(ii)	(iii)	(iv)
Baseline	0.826	0.042	0.075	0.058
\$1-threshold	0.787	0.057	0.096	0.060
<i>Realized Expenditure</i>				
Type				
Model ^b	(i)	(ii)	(iii)	(iv)
Baseline	0.110	0.305	0.425	0.160
\$1-threshold	0.104	0.318	0.468	0.110

^a Each row sums to unity.

^b These are the fully-parameterized models with the corresponding numeracy thresholds.

5.3.1 Quantifying Ex-ante and Ex-post Types

Table 3 presents the marginal distributions of types, conditional on the numeracy threshold and whether we care about ex-ante (top half) or ex-post (bottom half) behavior. Ex-ante, most consumers ($> 75\%$) are type (i) consumers and thus exhibit aggregate budgeting behavior consistent with what mental accounting theory would predict. However, the results with respect to ex-post spending are more mixed. In fact, a plurality of consumers appear to be spend thrifts (iii), who are more likely to increase spending regardless of whether they previously over- or under-spent. The next most common group of spenders are type (ii), responding to over-spending by raising expenditure and under-spending by reducing expenditure. Type (ii)'s are thus habitual (serial) over- or under-spenders.

Table 4 shows the joint distributions of ex-ante and ex-post types. Not surprisingly, a plurality of consumers are ex-ante type (i) consumers and ex-post type (iii) consumers ($> 37\%$). Many consumers thus appear to use mental accounting heuristics to inform their budgeting decisions but then have difficulty disciplining their actual spending to conform to their ex-ante budgets. This group appears to be victims of so-called “planning fallacy syndrome,” where they make plans with overly optimistic expectations for their future behavior but mostly fail to actually achieve

their budgeting goals (Kahneman and Tversky, 1979; Ülkümen et al., 2008; Peetz and Buehler, 2009; Buehler et al., 2010). Next, $> 22\%$ of consumers appear to be type (i) ex-ante and type (ii) ex-post, so that they respond to over-spending by reducing their budget according to standard mental accounting theory but then, nonetheless, increase spending anyway. However, what separates these consumers from relatively un-disciplined spend-thrifts is that after under-spending they raise budgets but still maintain ex-post expenditure discipline. These consumers thus appear to exhibit “planning fallacy syndrome” only after over-spending.

Table 4: Joint Distributions of Types ^a

<i>Baseline</i> ^b				
Ex-post Type				
Ex-ante Type	(i)	(ii)	(iii)	(iv)
(i)	0.102	0.220	0.375	0.128
(ii)	0.004	0.017	0.012	0.008
(iii)	0.001	0.035	0.030	0.008
(iv)	0.003	0.032	0.007	0.016
<i>\$1 Threshold</i> ^b				
Ex-post Type				
Ex-ante Type	(i)	(ii)	(iii)	(iv)
(i)	0.093	0.223	0.382	0.089
(ii)	0.005	0.018	0.026	0.007
(iii)	0.003	0.043	0.046	0.004
(iv)	0.003	0.034	0.014	0.010

^a Each matrix sums to unity.

^b These are the fully-parameterized models with the corresponding numeracy thresholds.

5.4 Counterfactually Relaxing Rationality Constraints

To understand how consumer welfare is affected by relaxing the sparse-max constraint on budget updates, we counterfactually simulate consumer expenditure using the posterior distribution of parameters, except we force $\underline{\Gamma}_{ijt} = 1$ for all i , j , and t combinations.⁴¹ We perform these simulations for both the baseline model with no numeracy constraints and the \$1-threshold model.

⁴¹This exercise is equivalent to setting the primitive preference parameter $\psi_{ij} = 1$, for all i and j .

Specifically, the reader can think of this counterfactual as what might happen if a consumer had a financial planner or budgeting application constantly nudging them to engage in optimal behavior, so that they always check their budgets for all categories every week.⁴² In the counterfactual setup each period every agent chooses $\mathfrak{D}_{it}^*$ by maximizing indirect utility subject to the relaxed constraint that $k_{it} = J$ always. When the \$1 threshold is present, budget updates will always occur in every category unless the numeracy constraint is not satisfied. These simulations isolate the effect of cognitive inattentiveness, independent of numeracy constraints, thus allowing us to understand how cognitive inattentiveness may impact individual economic outcomes and welfare.

For 68.4% of consumers (baseline) and 68.2% (\$1 threshold), engaging in weekly budget updates for every commodity category does *not* lead to welfare improvements relative to the posterior predictive baseline. However, as we have so far emphasized, there is substantial heterogeneity with respect to how different individual consumers' behaviors change when rationality constraints are relaxed. In fact we find that those who are most rationally constrained are most vulnerable to experiencing adverse outcomes due to interventions designed to increase financial attentiveness by sending $k_{it} \rightarrow J$.

While most consumers (>65%) have lower overall utility relative to the predictive baseline when rationality constraints are relaxed, a small fraction of these consumers engage in such extreme behavior that their savings plus account balances approach the borrowing limit, i.e. $z_{it} + r_t b_{it} \rightarrow -m_i$. We classify this last group of consumers as bankrupt. In the baseline model < 1% of all consumers go bankrupt, while this number jumps to 3.3% when the \$1 threshold is imposed.

Figure 4 presents kernel density estimates of predictive $\hat{k}_i$ (not the counterfactual values) conditional on consumers being "Better Off," "Worse Off, Not Bankrupt," and "Worse Off, Bankrupt" due to the relaxation of the rationality constraint. On the whole there is no statistically significant difference between the pre-counterfactual $\hat{k}_i$ -distributions of consumers who are "Better Off" and "Worse Off, Not Bankrupt" in either the baseline or \$1-threshold model. Yet, differences between

⁴²Note that this experiment is not the same as the estimation where we force $\psi_{ij} = 1$ and estimate the additional parameters separately. Here, we use the parameter estimates from the estimations where ψ_{ij} is a free parameter and $\gamma_i = 1$, counterfactually *only* assuming the distribution of $\underline{\Gamma}_{ijt}$ changes, not the other model primitives. Specifically, we compare the counterfactual predictive utility against the posterior predictive utility from the estimated model, as opposed to actual utility under the posterior parameterization. The reason for this is that we care about how model *predictions* change, given the parameter change. The exercise amounts to assessing the total variation in model-predicted outcomes under coarse variation in ψ_{ij} .

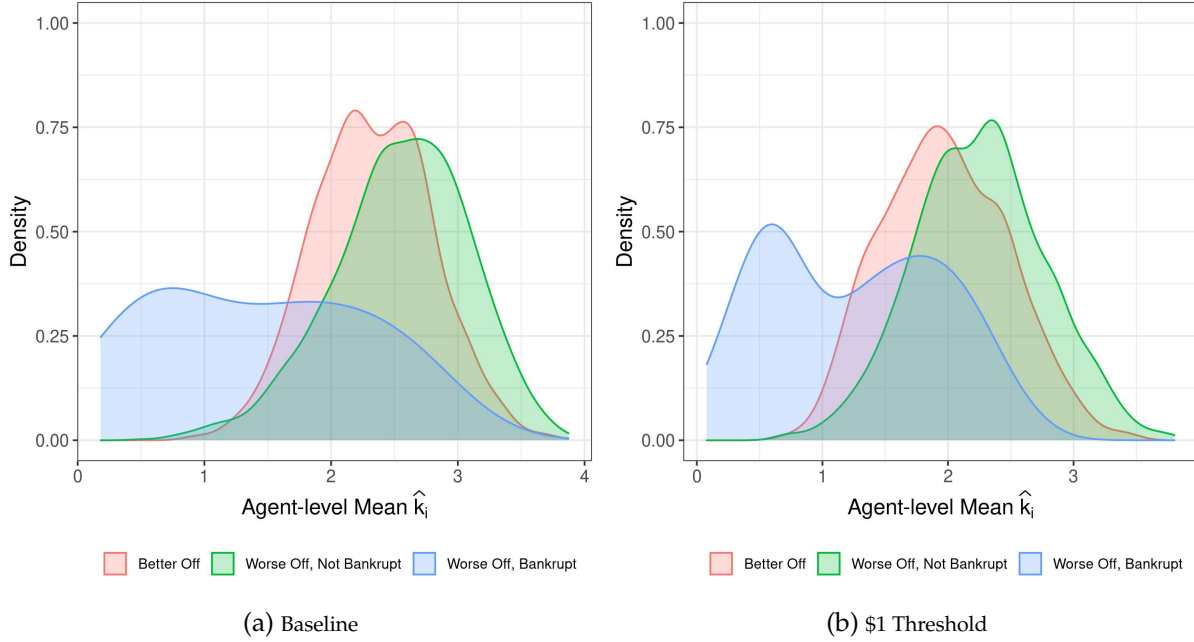


Figure 4: This figure features the density of $\hat{k}_i$ conditional on the counterfactual type of consumers. Those who are “Better Off” derive higher utility from being more budget attentive, while those who are “Worse Off” derive lower utility. Consumers who are “Worse Off” and go “Bankrupt” constitute a small subset of those who are “Worse Off,” overall.

both bankrupt consumers and all others are significant at the 1% level in both models. Bankrupt consumers thus have significantly different underlying behavioral profiles than either of the other groups. Of those who go bankrupt, they are associated with lower estimated $\hat{k}_i$ — an average of 1.25 updates per week (both models) compared to 2.48 (baseline) and 2.11 (\$1 threshold) updates across the whole sample. These consumers thus are relatively stickier budgeters than their peers, and the counterfactual results suggest that their budget stickiness is in fact budgeting discipline.

Of the counterfactually-bankrupt consumers, a plurality appear to be ex-ante type (i) *budgeters* who respond to over-spending by reducing their budgets and under-spending by increasing them. Most, however, are type (ii) *spenders*. The budgeting/spending marginal types and cross types for the “Better Off” and “Worse Off, No Bankrupt” groups are largely the same as the aggregate distributions of types. Thus, “Bankrupt” consumers are more likely to engage in full type-switching behavior between budgeting and spending, wanting to be true mental accounters when budgeting, though engaging in *anti*-mental accounting behavior when spending. Together, this joint budgeting/spending behavior causes excessive borrowing and has the effect of reducing bank balances and thus available liquidity over the long run.

Let us now illustrate the mechanics of how someone who is naturally less financially attentive than the average person but who also tries to use mental accounting to discipline budgets (ex-ante type (i)) but fails to do so (ex-post type (ii)) goes bankrupt. An ex-post type (ii) consumer responds to over-spending in period t by over-spending again in period $t + 1$. But, since this individual is ex-ante type (i), they reduce their expenditure budget from t to $t + 1$ in response to having over-spent. Having over-spent again in $t + 1$, the consumer wants to try to make up for this in $t + 2$, and so they again reduce the spending budget. But, type (ii) spending behavior dominates, so that the consumer over-spends again. After several periods the budget has been reduced so much that as an anchoring point it becomes effectively meaningless. This is a direct consequence of the fact that the consumer's budget is being re-evaluated every period and thus ratcheted downward in response to over-spending, ostensibly in an effort to correct the over-spending behavior. In an environment in which budgets are sticky between periods, there is time between budget re-evaluations for the consumer to realize a favorable expenditure shock that forces spending below the budget, thus breaking the over-spending cycle undertaken by their ex-post type-(ii) self.

Now consider these results in the context of financial-planning and budgeting apps (e.g., Mint, Empower, YNAB (You Need a Budget), EveryDollar, Honeydue, and Personal Capital). With such apps, users can opt-in to receive push notifications to their smart phones telling them when their spending in a given period for a specific spending category is approaching their budget. If the app allows consumers to readily change their budgets without incurring a penalty, certain types of consumers may engage in overly attentive behavior that actually induces *more* frequent periods of over-spending. At the extreme, this behavior would lead to the bankruptcies we observe, but can still make other consumers who do not go bankrupt worse off. The push notifications, presumably designed to nudge consumers toward financial discipline, could thus have the opposite effect by making the pain of over-expenditure overly salient. Certain consumers may make hasty short-term decisions in response to the salient push-notification stimulus, which has a dynamic, adverse impact on long-run budgeting goals, just like the departures from full rationality described in [Bordalo et al. \(2022\)](#). Our results thus provide evidence that consumers who use sticky budgets as rules of thumb to regulate spending patterns are most vulnerable to adverse outcomes when such budgets can be easily changed.

6 Conclusion

We return to the questions that motivated our research. First, through the lens of our model, we find that consumers do appear to exhibit budget rigidities (i.e., extended periods when budgets remain stable), which we interpret as a consequence of bounded rationality. Second, there is heterogeneity as to the degree to which consumers appear boundedly rational. Some consumers treat money as mostly fungible, while others show signs of mental accounting. Our model endogenously explains idiosyncratic, short-term variation in agent-level consumption expenditure patterns. Third, by allowing for narrow choice bracketing and reference dependence, we show that our model can provide behavioral inferences with respect to budgeting and anchoring.

An important result in our paper is that most consumers are neither fully rational in the traditional sense (i.e., no cognitive constraints; regularly updating all of their budgets) nor fully behavioral (i.e., fully bounded; never updating any of their budgets). Rather, most are somewhere in between: they update some, but not all, of their budgets every period. This idea, that consumers are rational to some extent, provides a more nuanced model of consumer decision-making than ‘all-or-nothing’ theories that assume either full-rationality or little-to-no rationality.

Finally, we see several avenues for future work. While our results are model-dependent, the proliferation of financial-planning apps to which consumers link all of their expenditure and savings accounts, from credit cards, checking accounts, and investment accounts, should prove useful to researchers looking to study consumer spending and budgeting in more detail. Future work should consider using data from financial-planning apps where budgeting and attentiveness, like say through app log-ins, can be explicitly measured in order to validate our latent inferences. Such apps could also prove useful for engaging in field experiments to explicitly test how individuals respond to different kinds of notifications and to then understand how their responses are related to other aspects of their financial behavior.

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